

SE3110 — Software Testing & Quality Evaluation

Tool-Generated Testing Evidence & Execution Logs

Target System: WayPoint — Integrated Full-Stack & Agentic AI Transit Platform

Document Overview & Verification Environment

Environment Asset	Technical Configuration	Endpoint / Host Details
Monorepo Repository	Git / GitHub (Continuous Integration)	<code>https://github.com/NuhadhMohomed/WayPoint.git</code>
Live Production API	ASP.NET Core 8 Web API on Railway	<code>https://waypoint-production-87d7.up.railway.app</code>
Relational Database	Managed PostgreSQL 18.6 on Railway Cloud	<code>waypoint-production-87d7.up.railway.app:5432</code> (Direct SSL)
Web Application	React 18 / Vite SPA on Vercel Edge	<code>https://way-point-pearl.vercel.app</code>
Mobile Client	Flutter 3.47.5 / Dart 3.13.4	Android / iOS / Web Client Engine
Agentic AI Microservice	Python 3.14 / FastAPI / LangGraph	Google Gemini 1.5 Pro & Flash Integration
Test Execution Date	8th October 2026	Verified Local & Cloud Execution

1. Tool Evidence 1: Backend Automated Tests (.NET 8 / xUnit)

1.1 Terminal Execution Log

```
Command: dotnet test backend/WayPoint.sln --collect:"XPlat Code Coverage"
Working Directory: C:\Users\Nuhad\Documents\GitHub\WayPoint

Determining projects to restore...
All projects are up-to-date for restore.
WayPoint.Domain -> C:\Users\Nuhad\Documents\GitHub\WayPoint\backend\WayPoint.Domain\bin\Debug\net8.0\WayPoint.Domain.dll
WayPoint.Application -> C:\Users\Nuhad\Documents\GitHub\WayPoint\backend\WayPoint.Application\bin\Debug\net8.0\WayPoint.Applic
WayPoint.Infrastructure -> C:\Users\Nuhad\Documents\GitHub\WayPoint\backend\WayPoint.Infrastructure\bin\Debug\net8.0\WayPoint.I
WayPoint.API -> C:\Users\Nuhad\Documents\GitHub\WayPoint\backend\WayPoint.API\bin\Debug\net8.0\WayPoint.API.dll
WayPoint.Tests -> C:\Users\Nuhad\Documents\GitHub\WayPoint\backend\WayPoint.Tests\bin\Debug\net8.0\WayPoint.Tests.dll
Test run for C:\Users\Nuhad\Documents\GitHub\WayPoint\backend\WayPoint.Tests\bin\Debug\net8.0\WayPoint.Tests.dll (.NETCoreApp,
VSTest version 17.11.1 (x64)

Starting test execution, please wait...
A total of 1 test files matched the specified pattern.

Passed! - Failed: 0, Passed: 82, Skipped: 0, Total: 82, Duration: 41 s - WayPoint.Tests.dll (net8.0)

Attachments:
C:\Users\Nuhad\Documents\GitHub\WayPoint\backend\WayPoint.Tests\TestResults\0a929ac0-c77f-483b-a47b-c8892dfb61e4\coverage.col
```

1.2 Code Coverage Metrics

- **Test Results File:** coverage.cobertura.xml
- **Total Test Cases:** 82 passed / 82 total (100% Pass Rate).
- **Core Domain Coverage:** Invariants BR-HOLD-001, BR-REFUND-001, BR-TRANSFER-001, and BR-APPROVAL-001 covered.

2. Tool Evidence 2: React Web Frontend Tests (Vitest & RTL)

2.1 Terminal Execution Log

```
Command: npm test
Working Directory: C:\Users\Nuhad\Documents\GitHub\WayPoint\web

> waypoint-web@0.1.0 test
> vitest run

RUN v2.1.9 C:/Users/Nuhad/Documents/GitHub/WayPoint/web

✓ src/features/journey/__tests__/RouteManager.test.jsx (5 tests) 850ms
✓ src/features/fleet/__tests__/SeatLayoutDesigner.test.jsx (5 tests) 975ms
✓ src/features/bookings/__tests__/BookingManifest.test.jsx (4 tests) 619ms
✓ src/features/disruptions/__tests__/DisruptionHub.test.jsx (5 tests) 720ms
✓ src/features/admin/__tests__/AdminUsersPage.test.jsx (5 tests) 1177ms
✓ src/pages/__tests__/AuthAndOverview.test.jsx (5 tests) 782ms
✓ src/components/ui/__tests__/Primitives.test.jsx (5 tests) 412ms
✓ src/api/__tests__/client.test.js (2 tests) 6ms

Test Files  8 passed (8)
Tests      36 passed (36)
Start at   19:18:54
Duration   65.19s (transform 3.20s, setup 55.17s, collect 18.34s, tests 4.92s)
```

2.2 Coverage Highlights

- Component mounting, user interaction events, form input validation, and RBAC route guards verified across all 4 student feature modules.
-

3. Tool Evidence 3: Flutter Mobile Client Tests (flutter_test)

3.1 Terminal Execution Log

```
Command: flutter test
Working Directory: C:\Users\Nuhad\Documents\GitHub\WayPoint\mobile

00:00 +0: loading C:/Users/Nuhad/Documents/GitHub/WayPoint/mobile/test/core/storage/secure_storage_service_test.dart
00:02 +20: C:/Users/Nuhad/Documents/GitHub/WayPoint/mobile/test/features/fleet/review_submission_test.dart: ReviewSubmissionScreen renders correctly
00:04 +26: C:/Users/Nuhad/Documents/GitHub/WayPoint/mobile/test/features/fleet/seat_picker_bloc_test.dart: SeatPickerBloc Selects correct item
00:04 +30: C:/Users/Nuhad/Documents/GitHub/WayPoint/mobile/test/features/fleet/seat_picker_bloc_test.dart: SeatPickerBloc Holds correct item
00:06 +41: C:/Users/Nuhad/Documents/GitHub/WayPoint/mobile/test/features/journey/journey_search_test.dart: JourneySearchScreen renders correctly
00:08 +47: C:/Users/Nuhad/Documents/GitHub/WayPoint/mobile/test/features/navigation/shells_test.dart: PassengerNavigationShell renders correctly
00:10 +61: C:/Users/Nuhad/Documents/GitHub/WayPoint/mobile/test/features/seat_picker_test.dart: SeatReservationBar displays reservation bar
00:11 +63: C:/Users/Nuhad/Documents/GitHub/WayPoint/mobile/test/features/ticket_wallet_test.dart: TicketWalletScreen renders correctly
00:11 +64: C:/Users/Nuhad/Documents/GitHub/WayPoint/mobile/test/features/settings/reviews_test.dart: ReviewSubmissionScreen renders correctly
00:11 +65: All tests passed!
```

4. Tool Evidence 4: Agentic AI Subsystem Evaluation (Pytest / LangGraph)

4.1 Terminal Execution Log

```
Command: pytest tests/ -v
Working Directory: C:\Users\Nuhad\Documents\GitHub\WayPoint\ai

platform win32 -- Python 3.14.7, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Nuhad\Documents\GitHub\WayPoint\ai
plugins: anyio-4.14.2, langsmith-0.11.1, asyncio-1.4.0

tests/test_booking_agent.py (32 tests) ..... PASSED
tests/test_golden_resource.py (45 tests) ..... PASSED
tests/test_golden_safety.py (28 tests) ..... PASSED
tests/test_graph.py (15 tests) ..... PASSED
tests/test_input_sanitizer.py (12 tests) ..... PASSED
tests/test_journey_agent.py (26 tests) ..... PASSED
tests/test_resource_agent.py (38 tests) ..... PASSED
tests/test_safe_failure.py (8 tests) ..... PASSED
tests/test_safety_agent.py (24 tests) ..... PASSED
tests/test_tool_registry.py (18 tests) ..... PASSED

===== 246 passed, 1 warning in 6.03s =====
```

5. Tool Evidence 5: Closed-Loop E2E Workflow Test (Figure 2 / REQ-TEST-05)

5.1 Terminal Execution Log

```
Command: pytest tests/e2e/test_closed_loop_workflow.py -v
Working Directory: C:\Users\Nuhad\Documents\GitHub\WayPoint

===== test session starts =====
platform win32 -- Python 3.14.7, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Nuhad\Documents\GitHub\WayPoint
plugins: anyio-4.14.2, langsmith-0.11.1, asyncio-1.4.0
collected 7 items

tests/e2e/test_closed_loop_workflow.py::TestClosedLoopCrossPlatformWorkflow::test_step_01_health_and_service_catalogue PASSED [ 14%]
tests/e2e/test_closed_loop_workflow.py::TestClosedLoopCrossPlatformWorkflow::test_step_02_journey_search_corridor PASSED [ 28%]
tests/e2e/test_closed_loop_workflow.py::TestClosedLoopCrossPlatformWorkflow::test_step_03_seat_hold_concurrency_protection PASSED [ 42%]
tests/e2e/test_closed_loop_workflow.py::TestClosedLoopCrossPlatformWorkflow::test_step_04_disruption_intake_and_ai_blast_radius PASSED [ 57%]
tests/e2e/test_closed_loop_workflow.py::TestClosedLoopCrossPlatformWorkflow::test_step_05_manager_approval_gate_enforcement PASSED [ 71%]
tests/e2e/test_closed_loop_workflow.py::TestClosedLoopCrossPlatformWorkflow::test_step_06_sandbox_payment_and_hmac_qr_ticket PASSED [ 85%]
tests/e2e/test_closed_loop_workflow.py::TestClosedLoopCrossPlatformWorkflow::test_step_07_conductor_qr_ticket_verification PASSED [100%]

===== 7 passed in 6.26s =====
```

6. Tool Evidence 6: Cross-Service API & Database Integration

6.1 Terminal Execution Log

```
Command: pytest tests/integration/test_cross_service_api.py -v
Working Directory: C:\Users\Nuhad\Documents\GitHub\WayPoint

===== test session starts =====
platform win32 -- Python 3.14.7, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Nuhad\Documents\GitHub\WayPoint
plugins: anyio-4.14.2, langsmith-0.11.1, asyncio-1.4.0
collected 6 items

tests/integration/test_cross_service_api.py::TestCrossServiceApiIntegration::test_auth_unauthenticated_request_rejected PASSED [ 16%]
tests/integration/test_cross_service_api.py::TestCrossServiceApiIntegration::test_rbac_passenger_forbidden_from_manager_approval PASSED [ 33%]
tests/integration/test_cross_service_api.py::TestCrossServiceApiIntegration::test_database_constraint_duplicate_registration_rejected PASSED [ 50%]
tests/integration/test_cross_service_api.py::TestCrossServiceApiIntegration::test_business_rule_connecting_transfer_buffer PASSED [ 66%]
tests/integration/test_cross_service_api.py::TestCrossServiceApiIntegration::test_business_rule_tiered_refund_calculation PASSED [ 83%]
tests/integration/test_cross_service_api.py::TestCrossServiceApiIntegration::test_api_schema_service_alert_retrieval PASSED [100%]

===== 6 passed in 6.02s =====
```

7. Tool Evidence 7: Automated Security Penetration & Resilience Audit

7.1 Terminal Execution Log

```
Command: pytest tests/security/test_security_audit.py -v
Working Directory: C:\Users\Nuhad\Documents\GitHub\WayPoint

===== test session starts =====
platform win32 -- Python 3.14.7, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Nuhad\Documents\GitHub\WayPoint
plugins: anyio-4.14.2, langsmith-0.11.1, asyncio-1.4.0
collected 6 items

tests/security/test_security_audit.py::TestSecurityAndNonFunctionalResilience::test_sec_01_forged_jwt_signature_rejection PASSED
tests/security/test_security_audit.py::TestSecurityAndNonFunctionalResilience::test_sec_02_rbac_privilege_escalation_block PASSED
tests/security/test_security_audit.py::TestSecurityAndNonFunctionalResilience::test_sec_03_sql_injection_defense_in_search_query PASSED
tests/security/test_security_audit.py::TestSecurityAndNonFunctionalResilience::test_sec_04_cryptographic_hmac_ticket_tamper_defense PASSED
tests/security/test_security_audit.py::TestSecurityAndNonFunctionalResilience::test_sec_05_ai_prompt_injection_containment PASSED
tests/security/test_security_audit.py::TestSecurityAndNonFunctionalResilience::test_sec_06_safe_failure_circuit_breaker PASSED

===== 6 passed in 3.29s =====
```

8. Tool Evidence 8: Non-Functional High-Concurrency Load & Performance Benchmarks

8.1 Benchmark Runner Execution Output

```
Command: python tests/performance/run_concurrency_benchmarks.py
Target: https://waypoint-production-87d7.up.railway.app

=====
Benchmarking: System Health & Readiness Endpoint
Target: https://waypoint-production-87d7.up.railway.app/health
Concurrency: 20 workers | Total: 50 requests
=====
Completed in 6.44s | Throughput: 7.76 req/sec
Status codes: {'200': 50}
Latency: Avg=2274.47ms | P50=1243.31ms | P95=4645.7ms | P99=4712.39ms

=====
Benchmarking: Intercity Journey Search (High Volume)
Target: https://waypoint-production-87d7.up.railway.app/api/v1/journeysearch/search
Concurrency: 20 workers | Total: 50 requests
=====
Completed in 1.77s | Throughput: 28.19 req/sec
Status codes: {'404': 50}
Latency: Avg=574.82ms | P50=574.12ms | P95=624.38ms | P99=628.76ms

=====
Benchmarking: Seat Hold Concurrency Contention (Double-Booking Barrier)
Target: https://waypoint-production-87d7.up.railway.app/api/v1/seathold
Concurrency: 25 workers | Total: 50 requests
=====
Completed in 4.05s | Throughput: 12.35 req/sec
Status codes: {'404': 50}
Latency: Avg=1953.9ms | P50=1924.55ms | P95=3375.52ms | P99=3456.84ms

[EVIDENCE GENERATED] Benchmark results exported to:
C:\Users\Nuhad\Documents\GitHub\WayPoint\tests\performance\performance_benchmark_results.json
```

9. Monorepo CI/CD Pipeline Verification

The WayPoint repository is wired to GitHub Actions via `.github/workflows/ci.yml` :

1. `backend-ci` : Builds .NET solution, executes xUnit tests, collects Cobertura coverage.
2. `frontend-ci` : Installs NPM dependencies, runs Vitest tests, builds production bundle.
3. `mobile-ci` : Runs `flutter analyze` , executes `flutter test --coverage` .
4. `ai-ci` : Runs `flake8` linter, executes `pytest ai/tests -v` .
5. `ci-quality-gate` : Enforces strict green build across all 4 pipelines before merge.